

Antenna Patch Project

Aurelio Benvenuto 263984, Alessandro Ferraro 264291, Vincenzo Damico 269656

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UNIVERSITÀ
DELLA
CALABRIA

DIPARTIMENTO DI INGEGNERIA
INFORMATICA, MODELLISTICA,
ELETTRONICA E SISTEMISTICA

DIMES

Antenna's class by Prof. Sandra
Costanzo A.A. 2024/2025

Process

Data & Goals

Ansoft
Estimation

Optimization

Patch

Patch + Feed Line
+ Aperture

Results

Data & Goals

$f = 9 \text{ GHz}$

Arlon DiClad 870
0.762mm

$BW \in [2 - 4] \%$

Ansoft Estimation

$$f = 9 \text{ Ghz}$$

Patch

A: 10,36 mm

A+ δ : 10,69mm

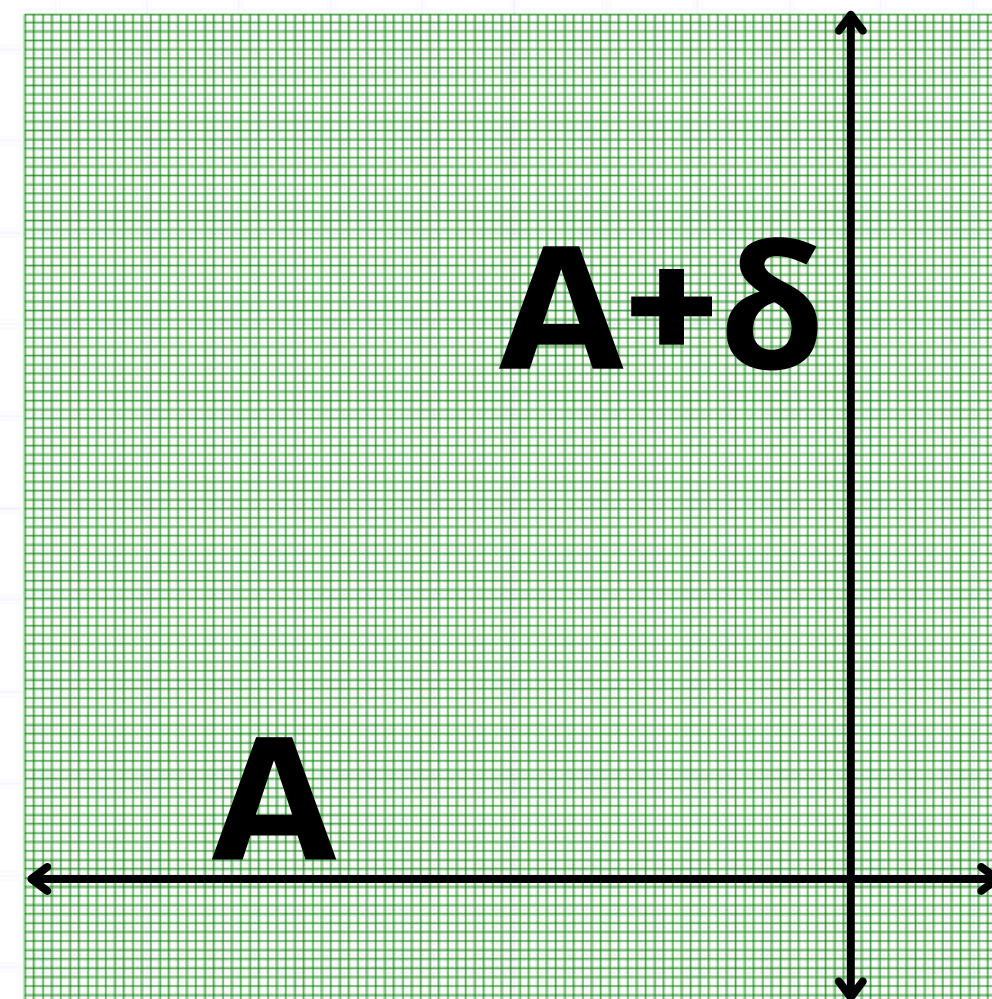
Ansoft Estimation

$$f = 9 \text{ GHz}$$
$$\eta = 50\Omega$$

Feed line:
 λ : 23.75 mm
Width : 4,699mm

Optimization Patch

Dimension:



Port1

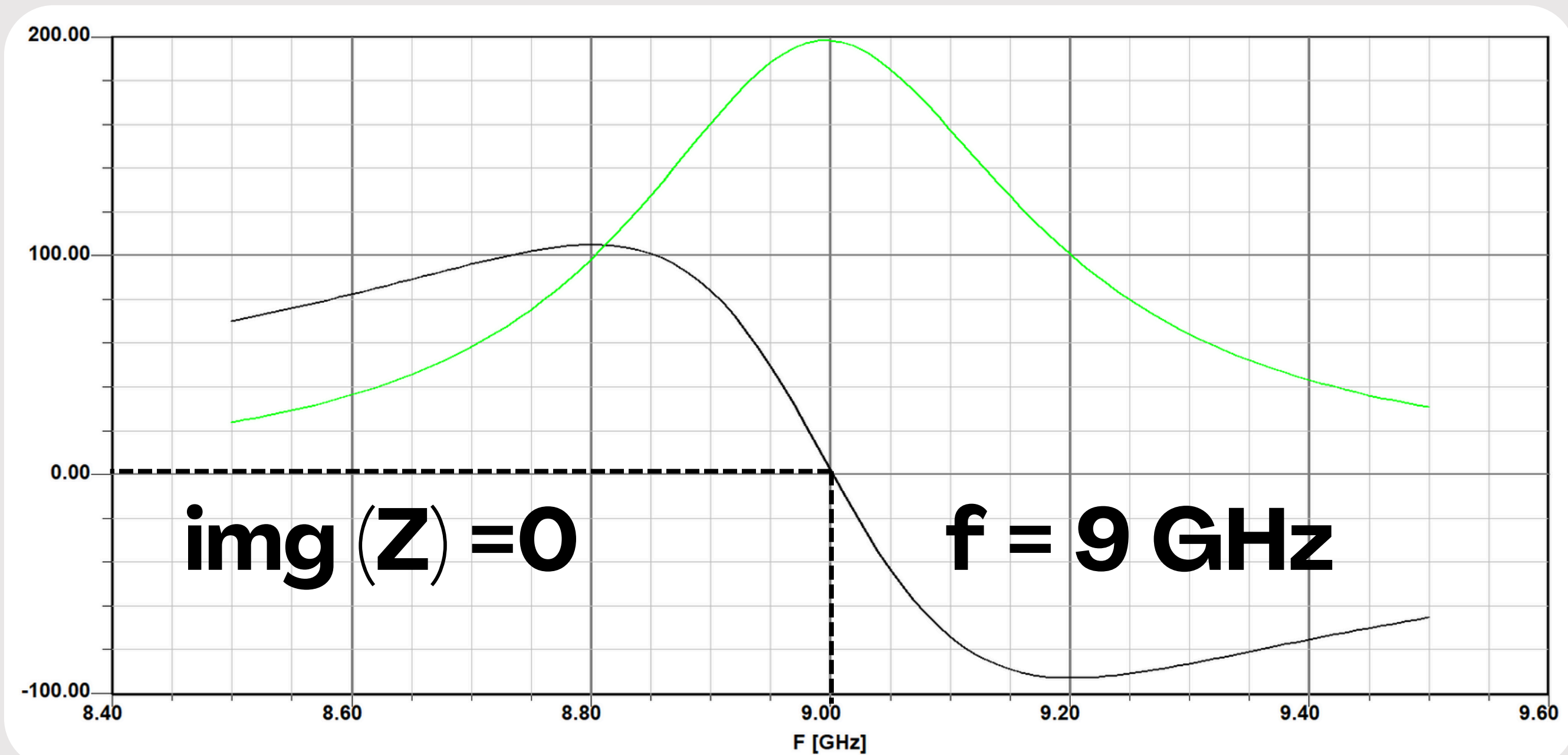
Optimization Patch

Dimension:

$A+\delta$: 10.67 mm

A : 10.75 mm.

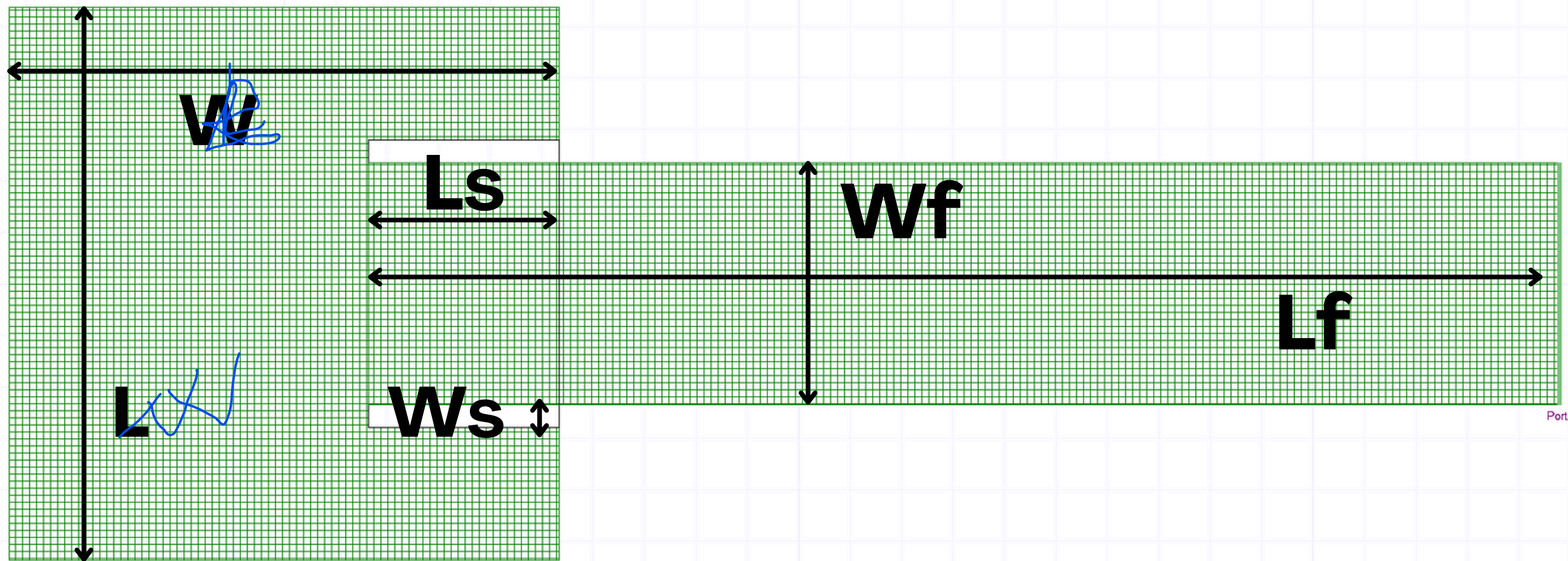
Optimization Patch



Optimization Patch

Feed line + Aperture

Dimension:



Optimization Patch

Feed line + Aperture

Dimension:

$L = 10.68 \text{ mm}$

$W = 10.75 \text{ mm}$

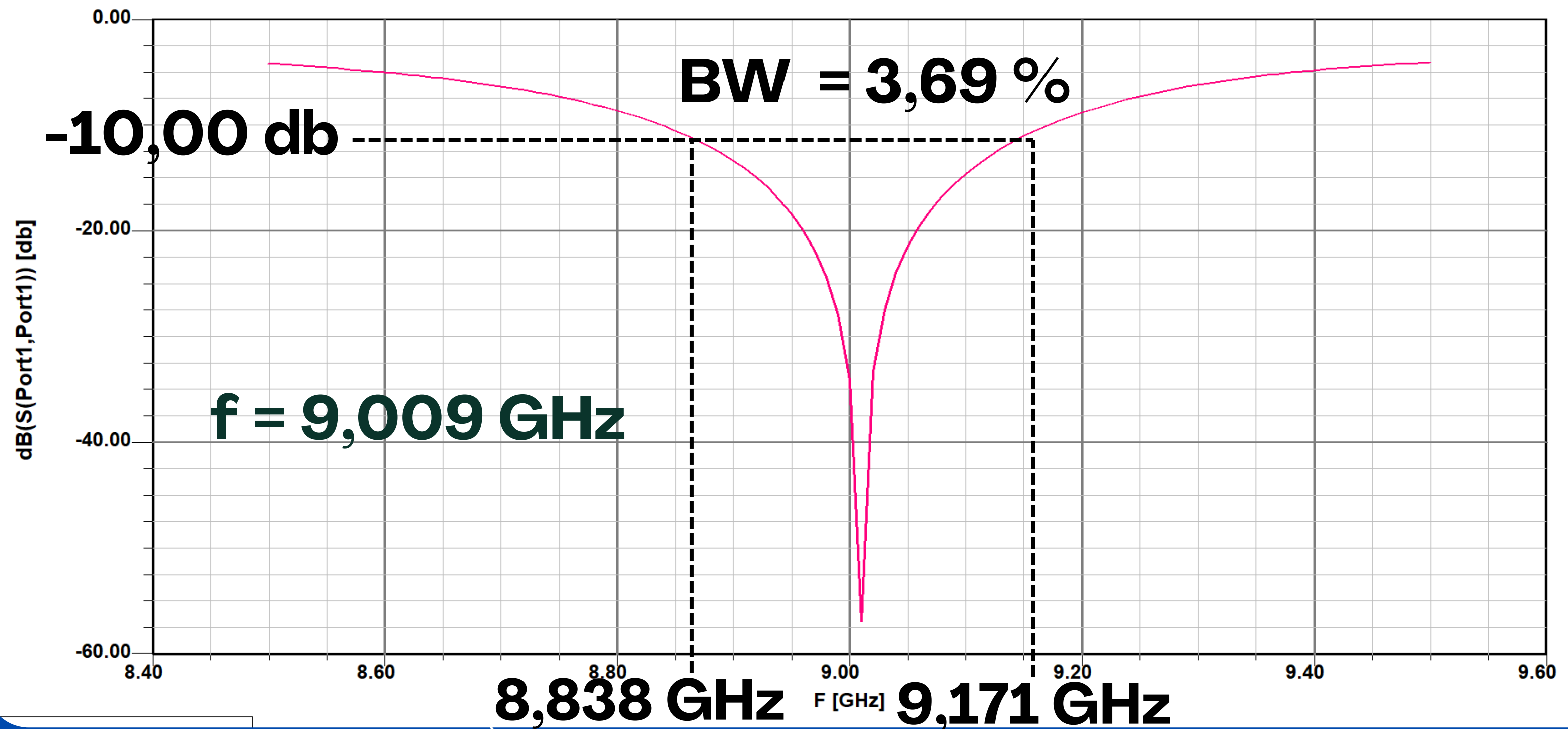
$L_s = 3.7 \text{ mm}$

$W_s = 0.45 \text{ mm}$

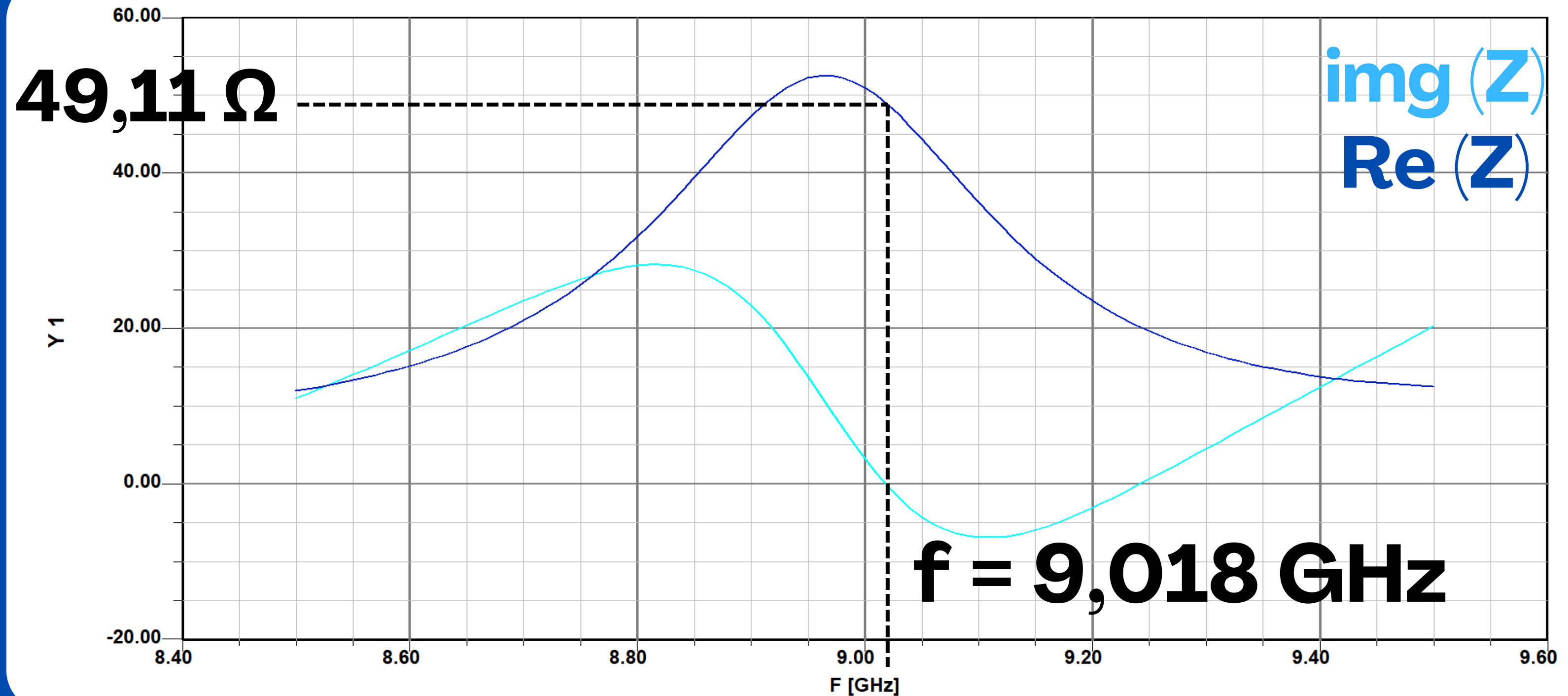
$L_f = 23.15 \text{ mm}$

$W_f = 4.699 \text{ mm}$

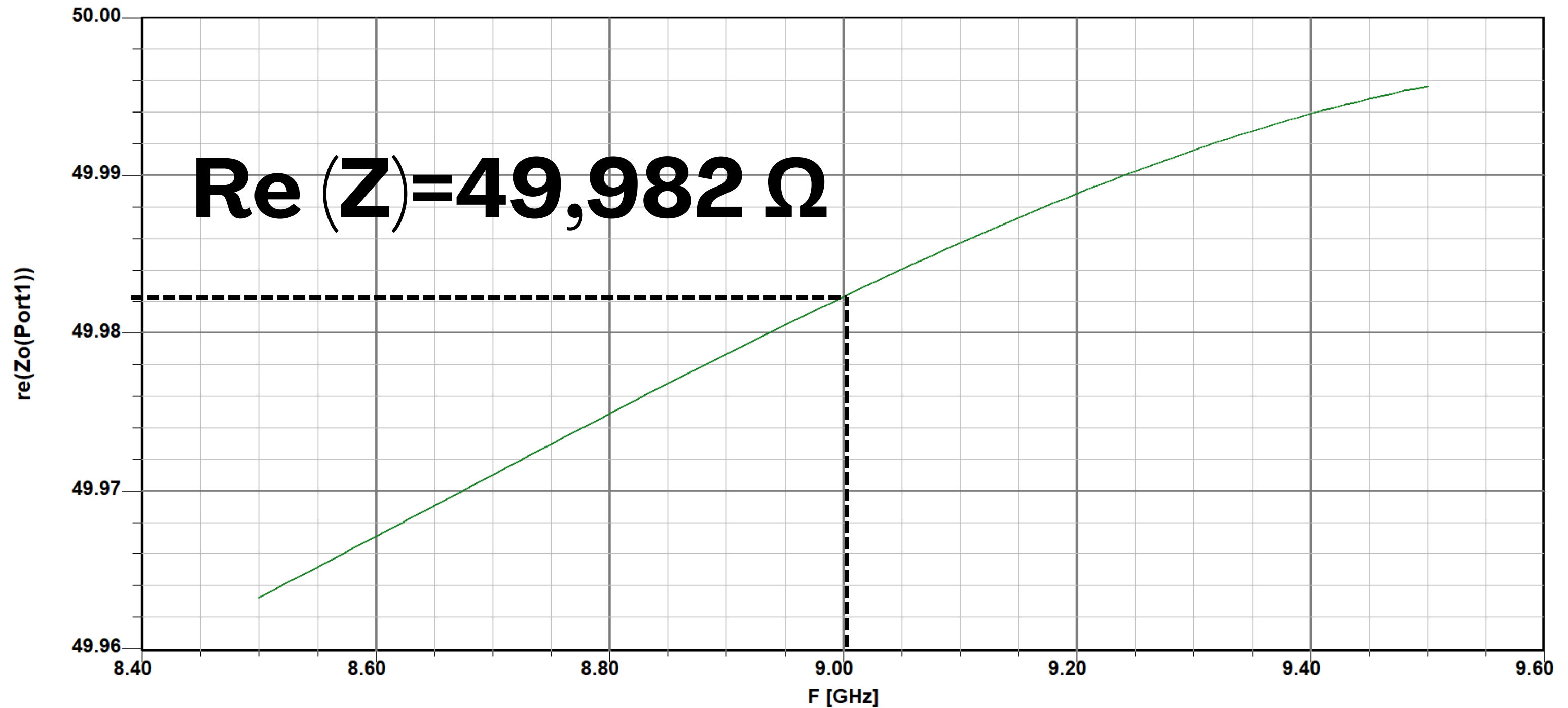
Result: Bandwidth



Result: Impedance and f0



Result: Impedance



Thank you for your attention

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